Vaibhav **KULKARNI**

Data & Bioinformatics Leader · Data Platforms, AI & Computational Biology

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Summary

Hands-on engineering leader with over a decade of experience building production-grade software, data platforms & machine-learning systems across life sciences, healthcare, supply chain & IoT, grounded in a PhD in Computer Science. At Debiopharm I architected the company's data fabric, from the central data repository to its scientific analytics platforms, & today lead data & bioinformatics across drug discovery & development: preclinical analytics, biomarker discovery, antibody developability & trial simulation. Self-taught computational biologist who builds & leads cross-functional teams of software & ML engineers, data scientists & bioinformaticians, & aligns them with scientific, quality & business stakeholders.

Experience

Debiopharm

Lausanne, Switzerland

Engineering Manager: Data & Bioinformatics — *July 2025 - Present*

- Lead a cross-functional team of data engineers, bioinformaticians & data scientists; set up a bioinformatics & data science service model for the Translational Medicine & Antibody Engineering teams, with a service catalogue, sprint-based intake & quarterly prioritisation
- Own preclinical analytics across in-vitro assays (cytotoxicity, internalization, binding) & in-vivo efficacy studies, delivering a dozen+ interactive dashboards used daily by translational scientists
- Lead transcriptomics biomarker discovery & validation for clinical-stage oncology programs: RNA-seq QC, batch correction, differential expression, pathway enrichment & ML gene signatures
- Deliver in-silico antibody developability & CMC risk assessment for antibody-drug conjugate (ADC) programs using protein language models, structure prediction & sequence-liability scanning
- Led the selection of & migration to a new NGS analysis platform, bringing omics processing in-house; built RNA-seq, variant-calling & ctDNA pipelines with automated, secure transfers from CROs & sequencing vendors
- Defined the preclinical data-flow & governance framework across the data repository, ELN & metadata systems, aligning Data Management, IT, Translational Medicine, Quality & wet-lab teams
- Initiated applied AI projects: generative models & Monte Carlo simulation for bioequivalence trial design, & a digital-pathology pipeline for IHC Ki-67 scoring

Data Engineering Team Lead — *July 2022 - June 2025*

- Architected & led, from scratch, the Central Data Repository: the company's hub for all clinical & non-clinical data, used by teams across several functions & all R&D programs
- Built the scientific analytics layer of the data fabric (SAS Viya on Azure Kubernetes, Posit suite, Terraform IaC), serving 200+ users & 100+ data products under GxP validation
- Delivered DropZone, a secure data-exchange platform for partners & vendors, in five months; used for in- & out-licensing due diligence
- Productionised privacy-aware LLM applications for document analysis & protocol writing, & an entity-recognition platform for scientific literature monitoring
- Automated GxP computerised system validation, infrastructure qualification & security audits through DevOps; led technical due diligence of healthcare start-ups for the corporate venture fund

SOPHiA GENETICS

Lausanne, Switzerland

Senior Data Engineer — *April 2021 - June 2022*

- Built terabyte-scale ETL pipelines turning raw genomics data into analytics-ready formats, with APIs & CLI tools that opened the data to internal teams
- Introduced regression testing for variant-detection pipelines & TLA+ formal specifications to verify the correctness of ETL operations

GenLots

Lausanne, Switzerland

Software Engineer | R&D Lead — *September 2019 - March 2021*

- Built a microservices platform delivering client ERP data to production ML services, including reinforcement-learning-based supply-chain planning, & led applied research with Swiss watch manufacturers

University of Lausanne

Lausanne, Switzerland

Research Scientist (PhD) — *November 2015 - August 2019*

- Led machine-learning & privacy research on large-scale mobility data (300+ participant study) & industry projects with public transport operators; 14 peer-reviewed publications

Earlier roles

Project Engineer, ETH Zürich (2015) · Embedded Systems Engineer, BOLT IoT (2011 - 2013)

Core Competencies

- **Data & Platforms:** Python, R, SQL, TypeScript, Rust · Azure, AWS, Kubernetes, Terraform · PostgreSQL, DuckDB, Airflow, dbt · SAS Viya, Posit, R Shiny
- **AI & ML:** PyTorch, scikit-learn, gradient boosting · protein language models (ESM-2, AbLang-2), structure prediction (AlphaFold, IgFold), omics foundation models (Geneformer) · generative models · LLM applications
- **Computational Biology:** RNA-seq, variant calling & ctDNA pipelines · biomarker discovery · antibody developability · in-vitro & in-vivo pharmacology analytics · PK / NCA · digital pathology
- **Leadership:** Building & leading cross-functional teams, technical strategy & roadmaps, vendor & platform selection, stakeholder management, GxP & quality, technical due diligence

Education

University of Lausanne *Lausanne, Switzerland*

PhD, Computer Science · Thesis: information systems for privacy-aware machine learning *2015 - 2019*

Eindhoven University of Technology *Eindhoven, Netherlands*

MSc, Embedded Systems, cum laude · Master thesis at ETH Zürich *2014 - 2015*

Technische Universität Berlin *Berlin, Germany*

MSc, Information & Communication Technology, with honours *2013 - 2014*

Selected Publications & Talks

- **Generative models for simulating mobility trajectories** · V. Kulkarni, N. Tagasovska, T. Vatter, B. Garbinato · *NeurIPS Workshop on Modeling & Decision-Making in the Spatiotemporal Domain*, 2018
- **Examining the limits of predictability of human mobility** · V. Kulkarni, A. Mahalunkar, B. Garbinato, J. D. Kelleher · *Entropy*, 2019
- **Breadcrumbs: a feature-rich mobility dataset with point of interest annotations** · A. Moro, V. Kulkarni, P. Ghiringhelli, B. Chapuis, K. Huguenin, B. Garbinato · *ACM SIGSPATIAL*, 2019
- **Information disclosure in location-based services: an extended privacy calculus model** · D. Naous, V. Kulkarni, C. Legner, B. Garbinato · *ICIS*, 2019
- **Talks:** Generalising ML models to unseen chemical spaces in drug discovery & Making informed technology choices for a clinical data repository (BioTechX Europe, 2023) · AI's role in the data space (Swiss Association of Actuaries, 2023)

14 peer-reviewed publications in total; full list on Google Scholar.

Awards, Learning & Beyond Work

- **Awards:** Swiss National Science Foundation grant for independent doctoral research (2019) · Zeno Karl Schindler award for master thesis, ETH Zürich (2015) · EIT ICT Labs award for best IoT business plan (2013)
- **Competitions:** Ginkgo Antibody Developability Competition (2025) · Kaggle Adaptive Immune Profiling Challenge (2025) · maintainer of awesome-biopharma-foundation-models
- **Certifications:** Agentic AI (Vanderbilt, 2026) · Finetuning LLMs (DeepLearning.AI, 2026) · ML in Production (DeepLearning.AI, 2025) · Drug Discovery (UC San Diego, 2024)
- **Beyond work:** Founded a STEM outreach initiative (25 rural schools, 400+ students) · endurance running (Lausanne Marathon, 20 km de Lausanne 2016 - 2025) · ultimate frisbee & padel